

INVESTIGATION 1 — NMAP PORT SCAN TRAFFIC ANALYSIS

Investigation Details	
Case ID	NTA-2026-001
Date & Time	June 4, 2026 — 00:40 AM UTC
Analyst	George Eze — SOC Analyst
Lab Series	Lab 4 — Network Traffic Analysis
Framework	NIST SP 800-61 Rev.2
Tool Used	Wireshark (Capturing from eth0)
Capture File	wireshark_eth0Y96PQ3.pcapng
Total Packets	14,376 packets captured
Displayed	1,006 packets (7.0%)
Attack Source	192.168.0.239 (Kali Linux)
Attack Target	192.168.0.138 (Ubuntu ICONIC)

Verdict	
X PORT SCAN CONFIRMED	
Attack Type	Nmap -sS SYN Scan
TCP Conversations	1,006 detected
SYN Packets	Rapid to all ports
Pre-Attack?	YES — Reconnaissance
Kill Chain Stage	Stage 1 — Recon
Breach Occurred	NO
Chrissie Method	APPLIED ■

EXECUTIVE SUMMARY

On **June 4, 2026 at 00:40 AM UTC**, a network traffic capture was initiated using **Wireshark on Kali Linux (eth0)** during a simulated SOC lab exercise. An **Nmap SYN scan (-sS)** was launched from **192.168.0.239 (Kali Linux)** against **192.168.0.138 (Ubuntu ICONIC)** — the monitored endpoint. Wireshark captured **14,376 total packets**, with **1,006 TCP conversations** detected using the filter **tcp.flags.syn==1 and tcp.flags.ack==0**. The capture revealed a classic **port scan signature** — one source IP generating SYN packets to multiple destination ports including 443, 8888, 143, 1720, 993, 587, 111, 80, 1025, 5900, 445 and 3389. This investigation directly applies **Chrissie's pre-attack timeline methodology** — demonstrating that reconnaissance activity occurred on the network **BEFORE** the main brute force attack was launched. This represents **Cyber Kill Chain Stage 1** (Reconnaissance) and maps to **MITRE ATT&CK; T1595** (Active Scanning) and **T1046** (Network Service Discovery). No breach occurred. Findings were used to build an early detection Suricata rule to catch this activity earlier in future incidents.

Wireshark Evidence		
Finding	Value	Significance
Total Packets Captured	14,376 packets	Full attack traffic captured in PCAP
Filter Applied	tcp.flags.syn==1 and tcp.flags.ack==0	Isolated SYN-only packets = port scan
Packets Displayed	1,006 (7.0%)	1,006 port scan packets identified
TCP Conversations	1,006 unique connections	One IP → 1,006 ports = confirmed scan
Source IP	192.168.0.239 (Kali Linux)	Attacker machine clearly identified
Target IP	192.168.0.138 (Ubuntu ICONIC)	Victim endpoint being enumerated
Ports Targeted	443, 8888, 143, 1720, 993, 587, 111, 80, 1025+	Attacker mapping all open services
Packet Size	58 bytes each	Consistent SYN packet size signature

Pre-Attack Timeline — Chrissie's Method		
Time	Event	Kill Chain
T-0:00	Nmap -sS launched from 192.168.0.239 against Ubuntu. Wireshark begins capture.	Stage 1 RECON
T-0:01	1,006 SYN packets sent to multiple ports — attacker mapping open services.	Stage 1 RECON
T-0:30	Nmap completes — attacker now knows port 22 (SSH) is open on target.	Stage 2 WEAPON
T-1:00	Hydra brute force launched specifically targeting port 22 — 14M passwords attempted.	Stage 3 DELIVERY
T-3:00	Wazuh detects 609 alerts. Brute force contained. Attacker IP blocked.	Stage 4 DETECTED
T+0:00	Pre-attack recon MISSED by Wazuh — new Suricata rule written to catch it next time.	LESSON LEARNED

Ports Discovered by Attacker		
Port	Service	Risk if Open
22	SSH	CRITICAL — Brute force target
80	HTTP	HIGH — Web attacks possible
443	HTTPS	MEDIUM — Encrypted traffic
445	SMB	CRITICAL — EternalBlue risk
3389	RDP	CRITICAL — Remote access
5900	VNC	HIGH — Remote desktop
993	IMAPS	MEDIUM — Email access
587	SMTP	MEDIUM — Email relay
143	IMAP	MEDIUM — Email access
8888	Alt HTTP	LOW — Non-standard port

MITRE ATT&CK Mapping			
ID	Technique	Tactic	Evidence
T1595	Active Scanning	Reconnaissance	1,006 SYN packets to all ports
T1046	Network Service Discovery	Discovery	Ports 22,80,443, 445,3389 found
T1110	Brute Force (Follow-on)	Credential Access	SSH port 22 targeted next
T1018	Remote System Discovery	Discovery	Ubuntu host enumerated

Indicators of Compromise (IOCs)		
Type	Value	Action
Attacker IP	192.168.0.239	BLOCK at firewall
Target IP	192.168.0.138	MONITOR closely
Protocol	TCP SYN scan	Alert on pattern
PCAP File	wireshark_eth0Y96PQ3	PRESERVE evidence
Open Port	22 (SSH)	HARDEN immediately
Open Port	445 (SMB)	RESTRICT access

Response Actions & Recommendations

#	Action	Priority	Status
1	Block 192.168.0.239 at firewall — attacker IP confirmed via Wireshark	CRITICAL	RECOMMENDED
2	Create Suricata rule to detect SYN scan (20+ SYN packets in 10 seconds)	HIGH	IMPLEMENTED
3	Restrict SSH access — disable password auth, use key-based only	HIGH	RECOMMENDED
4	Close unnecessary open ports — especially 445 (SMB) and 5900 (VNC)	HIGH	RECOMMENDED
5	Add port scan alert to Wazuh dashboard for real-time SOC visibility	MEDIUM	RECOMMENDED
6	Preserve PCAP file as evidence — store securely with chain of custody	MEDIUM	DONE
7	Add pre-attack recon detection to standard IR playbook	MEDIUM	PROCESS UPDATE

Lessons Learned	
Reconnaissance Is Invisible Without Packet Capture	Wazuh SIEM detected the brute force attack but MISSED the Nmap reconnaissance that preceded it. Without Wireshark packet capture, this attack would have gone undetected.
1,006 Ports in Seconds = Clear Attacker Signature	The Conversations window showing 1,006 TCP connections from a single IP in seconds is an unmistakable port scan signature. Real-time monitoring of this pattern is critical for early detection.
New Suricata Rule Created — Early Detection Now Active	Based on this investigation, a new Suricata rule was written to alert on 20+ SYN packets from a single source in 10 seconds. This move significantly improved our detection capabilities.

References: NIST SP 800-61 Rev.2 · MITRE ATT&CK; T1595/T1046 · Wireshark · Nmap · Suricata · Wazuh 4.10.3	Analyst: George Eze GitHub: GeorgeEzeSOCAnalyst Mentor: Chrissie (TS Academy) Date: June 4, 2026
---	--